

Computability & Logic, Exercises to Lectures 15+16

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Theoretical Exercises

From John C. Martin book

1 Standard Material

Push Down Automata

- 5.5 (b,c) [Just draw PDAs as diagrams, instead of writing transition tables]
- 5.6 (b)

Deterministic PDA

- 5.18 (a)

Pumping Lemma for Context-Free Languages

- 6.2 (d)
- 6.5 (b)

Closure Properties of (Deterministic) Context-Free Languages

- 6.12 (b,c) (for CFL)
- 6.20 (a) (for DCFL)

2 Bonus Material

Extra Exercise Material

- 5.28(a) CFG to PDA (useful)
- 5.8 (b) (more PDA exercise material)
- 5.18(b,d) (more DPDA exercise material)
- 6.5(d) (one more pumping lemma)
- 5.12 (prove something on special PDA)
- 6.20 (c) Closure properties of DCFL
- 5.24 (First read the proof of Theorem 5.16, not treated in detail)
- 6.11 (b) Ogden's Lemma (Not treated in lecture: First read Theorem 6.7)